

Sydney Girls High School



2012
MATHEMATICS EXTENSION 2
 [November 2011]

YEAR 12
ASSESSMENT TASK 1

Time Allowed: 60 minutes + 5 minutes reading time.

Name: _____

Teacher: _____

Topics: Uniform Circular Motion, Curve Sketching.

General Instructions:

- There are EIGHT (8) Questions which are not of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number clearly at the top of each question and clearly number each question.
- Use $g = 10\text{ms}^{-2}$ in all questions.

Total: 50 marks

Question 1 (6 marks)

Given $f(x) = x^2 - 1$ draw sketch graphs of the following:

(a) $y = f(x)$

1

(b) $y = -f(x)$

1

(c) $y = \frac{1}{f(x)}$

2

(d) $y^2 = f(x)$

2

Question 2 (8 marks)

Sketch the following showing any intercepts or asymptotes:

(a) $9y^2 + 25x^2 = 1$

2

(b) $(x+3)(y+2) = 4$

2

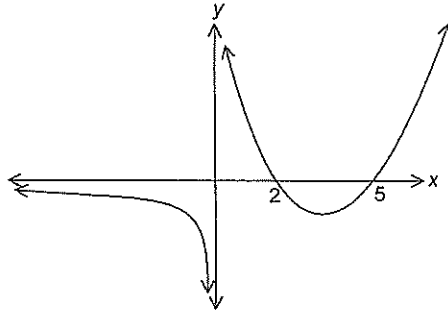
(c) $x^2 - 9y^2 = 1$

2

(d) $y = \frac{1}{(x-2)(x+1)^2}$

2

Question 3 (10 marks)



The diagram is a sketch of the function $y = f(x)$ which cuts the x axis at $x = 2$ and $x = 5$.

The graph has asymptotes $x = 0$ and $y = 0$. On separate diagrams sketch:

- | | |
|------------------------|---|
| (a) $y = f(x) $ | 1 |
| (b) $y = f(x)$ | 1 |
| (c) $y = [f(x)]^2$ | 2 |
| (d) $2^{f(x)}$ | 2 |
| (e) $y = f(x^2)$ | 2 |
| (f) $y = \sqrt{f(-x)}$ | 2 |

Question 4 (3 marks)

A girl ties a stone of mass 0.3 kg to the end of a piece of string, then swings it in a circle of radius 1.5 m . If the tension in the string is 3 N , calculate:

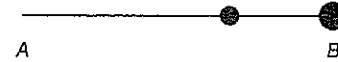
- | | |
|---------------------------------|---|
| (a) the speed of the stone | 1 |
| (b) the frequency of its motion | 1 |
| (c) its angular velocity | 1 |

[2]

Question 5 (4 marks)

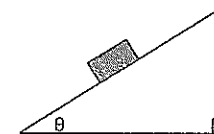
A 5 metre piece of string AB has a mass of 4 kg attached at point B . The string is then rotated about A in a horizontal circle. If the string breaks when the speed of rotation reaches 15 rad/sec ,

- | | |
|--|---|
| (a) Find the breaking strain of the string | 1 |
| (b) If the 4 kg mass at B is replaced by a 3 kg mass at the same position, and a 1 kg mass 2 metres from A , find the new maximum speed of rotation (in rad/sec) | 3 |



Question 6 (7 marks)

A railway track goes around a curve of radius 500 m . The track is inclined at θ° to the horizontal. The forces acting on a train taking the curve are the weight force mg , a sideways frictional force F and the normal reaction force.

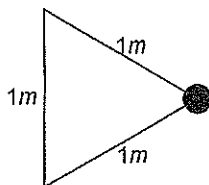


- | | |
|---|---|
| (a) Draw a diagram of the track showing all forces acting on a train taking the corner. | 1 |
| (b) By resolving forces vertically and horizontally derive an expression for F that is independent of N . | 3 |
| (c) At what angle should the track be banked for a designed train speed of 20 ms^{-1} ? | 2 |

[3]

Question 7 (7 marks)

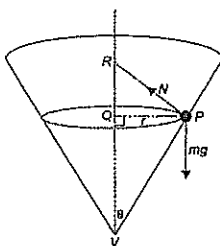
A 1 kg ball is attached to a rigid vertical rod by means of two strings each one metre long. The strings are attached to the rod one metre apart. The system is rotating about the rod at ω radians per second. Both strings are taut and form an equilateral triangle with the rod.



- (a) What is the tension in the lower string (in terms of ω)? 4
- (b) What is the speed (v) of the ball, if the ratio of the tension in the upper string to the tension in the lower string is 3:1? 3

Question 8 (6 marks)

A smooth hollow cone of semi vertex angle θ is placed with axis vertical and vertex V downwards. A particle P moves in a circle on its inner surface making x revolutions per second.



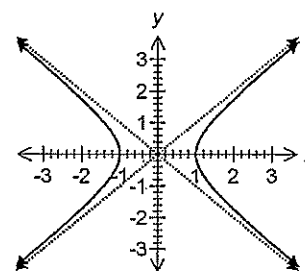
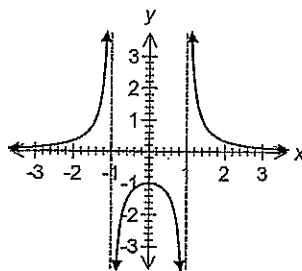
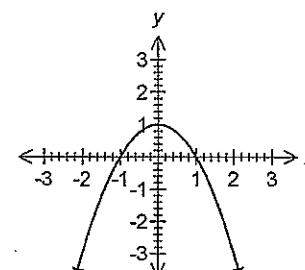
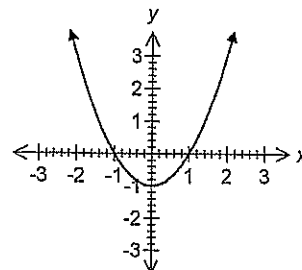
- (a) Show that the distance (r) of the particle from the axis of the cone at any time is: $\frac{g \cot \theta}{4\pi^2 x^2}$. 2
- (b) If $\theta = 30^\circ$ and $x = 1$ find VQ . 2
- (c) If the linear velocity is to be doubled, find the minimum height of the cone 2

End of paper.

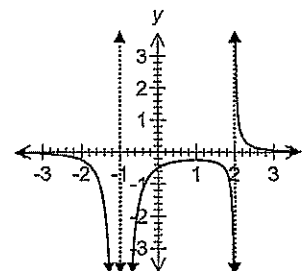
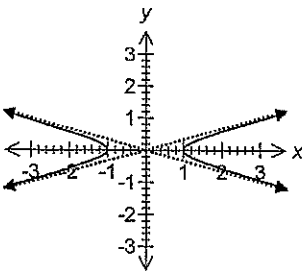
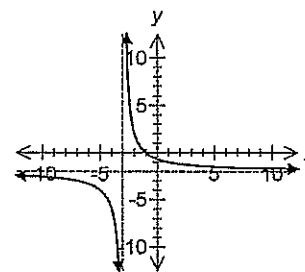
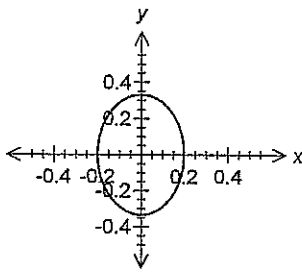
Marks

Extension 2 Mathematics Task 1 2012 Solutions

Question 1

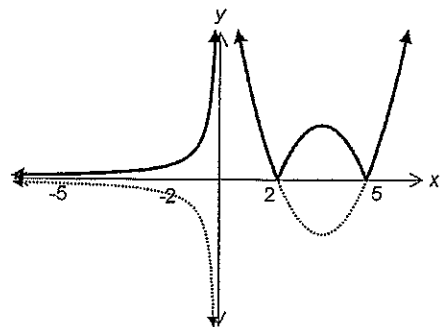


Question 2

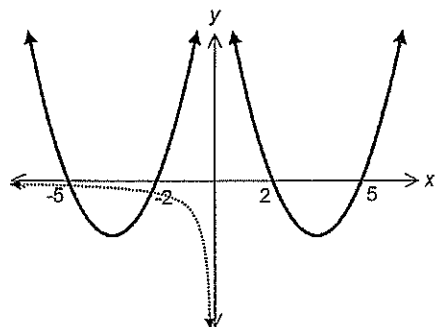


Question 3

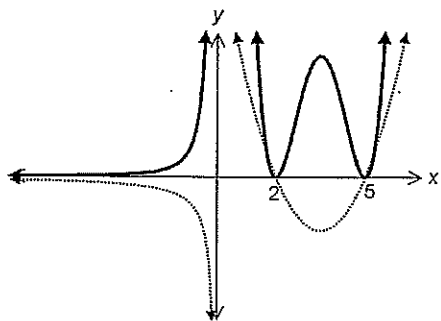
a.



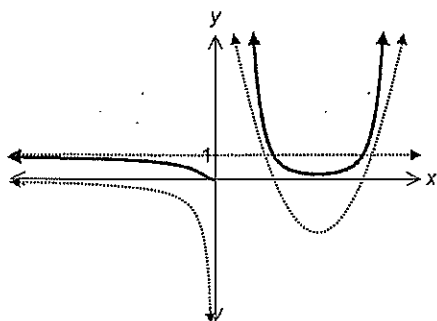
b.



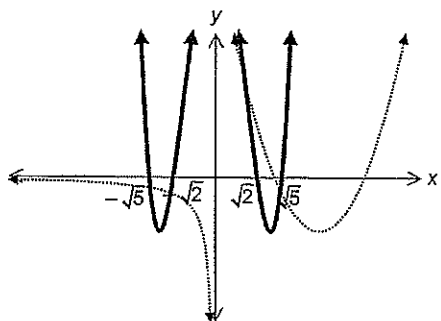
c.



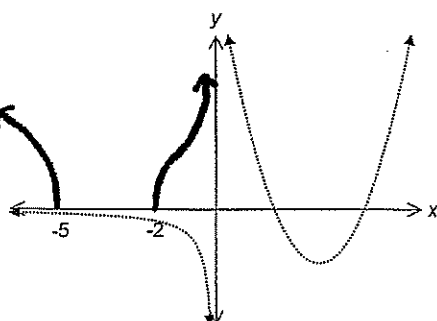
d.



e.



f.



Question 4

a. $T = \frac{mv^2}{r}$

$$3 = \frac{0.3 \times v^2}{1.5}$$

$$v^2 = 15$$

$$v = \sqrt{15} \text{ ms}^{-1}$$

b. $v = r\omega$

$$\sqrt{15} = 1.5\omega$$

$$\omega = \frac{\sqrt{15}}{1.5}$$

$$= 2.58 \text{ rad/sec}$$

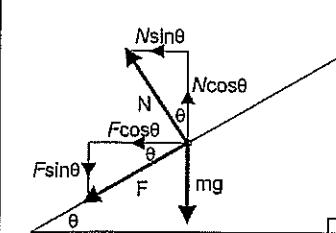
c. $f = \frac{\omega}{2\pi}$

$$= \frac{2.58}{2\pi}$$

$$= 0.41$$

Question 6

a.



b.

Horizontally:

$$F \cos \theta + N \sin \theta = \frac{mv^2}{r} \rightarrow (1)$$

Vertically:

$$N \cos \theta = F \sin \theta + mg$$

$$N \cos \theta - F \sin \theta = mg \rightarrow (2)$$

$$(1) \times \cos \theta$$

$$F \cos^2 \theta + N \cos \theta \sin \theta = \frac{mv^2}{r} \cos \theta \rightarrow (3)$$

$$(2) \times \sin \theta$$

$$N \sin \theta \cos \theta - F \sin^2 \theta = mg \sin \theta \rightarrow (4)$$

Question 5

a. $T = mr\omega^2$

$$= 5 \times 3 \times 15^2$$

$$= 3375 \text{ N}$$

b. $T_{bx} = mr\omega^2$

$$= 3 \times 3 \times \omega^2$$

$$= 9\omega^2$$

$$T_{ax} = 9\omega^2 + 1 \times 2 \times \omega^2$$

$$= 11\omega^2$$

$$3375 = 11\omega^2$$

$$\omega^2 = \frac{3375}{11}$$

$$\omega = 5.28$$

(3) - (4)

$$F = \frac{mv^2}{r} \cos \theta - mg \sin \theta \rightarrow (5)$$

c.

Substitute:

$$F = 0$$

$$v = 20 \text{ ms}^{-1}$$

$$r = 500$$

$$0 = \frac{m \times 20^2}{500} \cos \theta - 10m \sin \theta$$

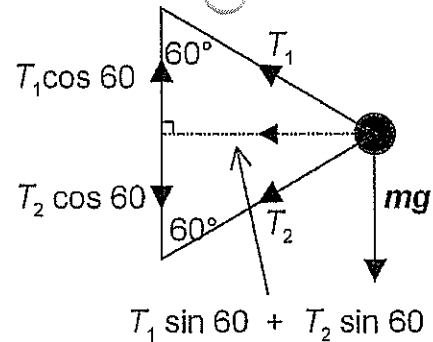
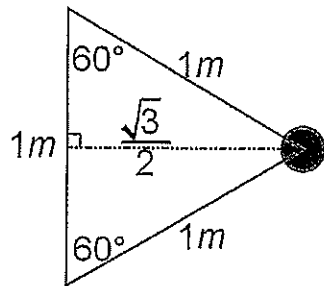
$$10m \sin \theta = \frac{m \times 20^2}{500} \cos \theta$$

$$\tan \theta = \frac{20^2}{500 \times 10}$$

$$\tan \theta = \frac{2}{25}$$

$$\theta = 4^\circ 34'$$

Question 7



a.

Horizontally:

$$\sin 60 (T_1 + T_2) = m r \omega^2 \rightarrow (1)$$

Vertically:

$$T_1 \cos 60 = T_2 \cos 60 + mg$$

$$\cos 60 (T_1 - T_2) = mg \rightarrow (2)$$

From (1)

$$\frac{\sqrt{3}}{2} (T_1 + T_2) = 1 \times \frac{\sqrt{3}}{2} \omega^2$$

$$T_1 + T_2 = \omega^2 \rightarrow (3)$$

From (2)

$$\frac{1}{2} (T_1 - T_2) = 10$$

$$T_1 - T_2 = 20 \rightarrow (4)$$

$$(3) - (4)$$

$$2T_2 = \omega^2 - 20$$

$$T_2 = \frac{\omega^2 - 20}{2}$$

b.

$$T_1 - \frac{\omega^2 - 20}{2} = 20$$

$$T_1 = \frac{20 + \omega^2}{2}$$

$$T_1 : T_2 = 3 : 1$$

$$T_1 = 3T_2$$

$$\frac{20 + \omega^2}{2} = 3 \left(\frac{\omega^2 - 20}{2} \right)$$

$$20 + \omega^2 = 3\omega^2 - 60$$

$$2\omega^2 = 80$$

$$\omega^2 = 40$$

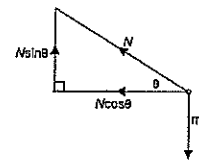
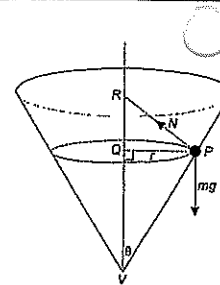
$$\omega = 2\sqrt{10} \text{ rad/sec}$$

$$v = r\omega$$

$$= \frac{\sqrt{3}}{2} \times 2\sqrt{10}$$

$$= \sqrt{30} \text{ ms}^{-1}$$

Question 8



a.

Horizontally:

$$N \cos \theta = m r \omega^2 \rightarrow (1)$$

Vertically:

$$N \sin \theta = mg \rightarrow (2)$$

$$(2) \div (1)$$

$$\tan \theta = \frac{g}{r \omega^2}$$

Substituting:

$$\omega = 2\pi x$$

$$\tan \theta = \frac{g}{r 4\pi^2 x^2}$$

$$r = \frac{g \cot \theta}{4\pi^2 x^2}$$

b.

If $\theta = 30^\circ$ and $x = 1$

$$r = \frac{10 \cot 30}{4\pi^2}$$

$$= 0.44 \text{ m}$$

$$\tan 30 = \frac{0.44}{h}$$

$$h = \frac{0.44}{\tan 30} = 0.76 \text{ m}$$

d.

$$v = 2 \times 0.44 \times 2\pi$$

$$= 1.76\pi$$

$$\frac{r}{h} = \frac{rg}{v^2}$$

$$h = \frac{v^2}{g} = 3.06 \text{ m}$$

